

Project Support Session

Agenda

- Project Problem Statement and Dataset Overview
- Open Q&A
- Project Evaluation Rubric
- Project Submission Guidelines
- FAQs

Project Problem Statement

Business Context

Lumina Energy Partners is a premier global investment firm managing a portfolio across traditional fossil fuels, renewable power generation, and advanced energy infrastructure. The global energy landscape is currently undergoing an unprecedented and highly complex transition. As the “Age of Electricity” takes hold, power demand is surging due to the electrification of transport and heating and the rapid expansion of AI-driven data centres. At the same time, the market is experiencing major structural shifts, including a coming wave of liquefied natural gas production, plateauing global oil demand, slower coal plant retirements due to policy support, and persistent bottlenecks in power grid infrastructure that constrain renewable integration.

To maximise portfolio returns and mitigate risk, Lumina’s investment committee requires rapid cross-sector intelligence. Evaluating an investment in a US natural gas facility, for example, requires analysts to synthesise insights spanning multiple energy markets. They must understand how AI data centres could account for almost half of projected US electricity demand growth and potentially add nearly 20 bcm of annual natural gas demand by 2030. At the same time, they must consider countervailing developments such as the slowdown in coal plant retirements due to federal policy support and the growing constraints on renewable deployment caused by grid congestion. These interdependencies require analysts to combine signals from multiple technical sources to produce a coherent macro outlook.

Currently, Lumina’s analysts rely on manual review of dense and highly technical International Energy Agency reports covering Coal, Oil, Gas, Electricity, and Renewables. Preparing a cross-sector investment briefing typically requires two to three days of work, during which analysts review five to ten separate reports and spend roughly 65% of their time locating relevant passages rather than performing financial modelling or scenario analysis. The process relies heavily on manual search across thousands of pages of PDFs and cross-referencing fragments of information between reports.

Project Problem Statement

Business Context

This workflow creates measurable operational inefficiencies and analytical risk. Citation traceability is inconsistent, subtle correlations between reports may be overlooked, and valuable analyst time is diverted away from high-value modelling and strategic evaluation. In fast-moving market conditions or competitive bidding windows, the delay inherent in manual synthesis can mean that investment opportunities are missed before a complete briefing is produced.

To remain competitive, Lumina requires an AI-assisted system capable of reducing briefing preparation time by at least 90%, retrieving relevant report passages with at least 90% accuracy, and generating fully grounded responses with complete source attribution and minimal hallucination risk. By transitioning from manual document review to a Retrieval Augmented Generation system grounded exclusively in authoritative IEA reports, Lumina can enable analysts to ask complex natural language questions and receive instant, synthesised, and fully cited strategic briefings. This approach would eliminate days of manual document review, reduce the risk of human oversight, and allow analysts to focus on financial modelling and strategic decision-making rather than time-consuming information retrieval.

Project Problem Statement

Objective

Your objective as a Data Scientist is to design, build, and evaluate a prototype Retrieval Augmented Generation (RAG) pipeline that enables Lumina's analysts to rapidly obtain accurate and fully traceable energy market insights from a corpus of dense, technical International Energy Agency market reports. The system must synthesise information across multiple reports while producing high-fidelity, hallucination-free responses suitable for high-stakes investment decision-making.

Key tasks include:

- **Data Ingestion and Preprocessing:** Extract text, quantitative data points, and complex policy details from the provided IEA PDF reports. Implement an intelligent chunking strategy that preserves the context of dense market analyses, outlook sections, and policy frameworks so that cross-sector insights can be retrieved accurately.
- **Vector Embedding and Storage:** Select an embedding model capable of capturing the semantic meaning of highly technical energy terminology and relationships across the reports. Generate semantic representations of the text and store these embeddings in a vector database to enable efficient similarity search across the entire corpus.
- **Advanced Retrieval:** Develop a robust retrieval mechanism capable of surfacing the most relevant passages across multiple reports in response to complex, multi-part analyst queries. The retrieval layer should support cross-sector questions that connect developments such as electricity demand growth, LNG supply expansion, and coal plant utilisation.
- **LLM Synthesis:** Integrate a Large Language Model to synthesise retrieved evidence into coherent, executive-level investment briefings. The generated responses must use the source reports for each key claim in order to minimise hallucination risk.
- **Evaluation:** Establish evaluation metrics to assess system performance against benchmark questions generated by Lumina's senior energy analysts. Evaluation should measure retrieval relevance and factual accuracy, to achieve approximately 90% retrieval accuracy and significantly reduce the time required to prepare cross-sector investment briefings.

Dataset Overview

- The corpus for this Retrieval-Augmented Generation (RAG) case study consists of five authoritative market reports published by the International Energy Agency (IEA). Each document provides a comprehensive analysis and medium-term forecasts up to the year 2030 for different energy sectors:

Document	Description
Coal 2025	- Analyses recent trends and provides forecasts through 2030 for global coal demand, supply, trade, and prices amid shifting energy policies. - IEA (2025), Coal 2025, IEA, Paris https://www.iea.org/reports/coal-2025, Licence: CC BY 4.0
Electricity 2026	- Examines global electricity systems, offering forecasts to 2030 on electricity demand, supply, grid infrastructure bottlenecks, and system flexibility. - IEA (2026), Electricity 2026, IEA, Paris https://www.iea.org/reports/electricity-2026, Licence: CC BY 4.0
Gas 2025	- Explores medium-term trends in global natural gas markets through 2030, with a specific focus on the impending global expansion of liquefied natural gas (LNG) capacity and low-emission gases. - IEA (2025), Gas 2025, IEA, Paris https://www.iea.org/reports/gas-2025, Licence: CC BY 4.0
Oil 2025	- Delivers detailed forecasts on evolving global oil supply, demand, refining, and trade dynamics through 2030, highlighting the rising influence of petrochemical feedstocks. - IEA (2025), Oil 2025, IEA, Paris https://www.iea.org/reports/oil-2025, Licence: CC BY 4.0
Renewables 2025	- Analyses recent policy and market developments to forecast the deployment of renewable energy technologies across the electricity, transport, and heat sectors up to 2030. - IEA (2025), Renewables 2025, IEA, Paris https://www.iea.org/reports/renewables-2025, Licence: CC BY 4.0

Open Q&A



Project Evaluation Rubric

Section	Points
Requirements Gathering	4
Answering Questions using the Base LLM	5
Answering Questions using the Base LLM with Prompt Engineering	7
Vector Database Setup for RAG	6
Answering Questions with the Base RAG System	7
Answering Questions with the Tuned RAG System	8
Output Evaluation	11
Business Insights & Recommendations	4
Presentation / Notebook - Overall Quality	8

Project Evaluation Rubric

Requirements Gathering

2

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9

What to Cover?

Business Problem Definition

Elaborate on the provided requirements

Guiding Questions and Considerations

Do your problem statement and analysis questions clearly align with the business objective, use data effectively, and aim to generate actionable insights?

Are the requirements broken down into measurable, data-driven steps that can be validated through analysis?

Project Evaluation Rubric

1	Answering Questions using the Base LLM	3	4	5	6	7	8	9
What to Cover? Show the answers to all the questions Provide comments/observations for all the answers received Guiding Questions and Considerations Define the response generation function. Use the function on all the given questions to generate the responses. How do the responses look like? Are they too generic?								

Project Evaluation Rubric

1	2	Answering Questions using the Base LLM with Prompt Engineering	4	5	6	7	8	9
What to Cover?		Guiding Questions and Considerations						
Mention and explain the engineered system prompt		What are the factors to consider when engineering a better prompt for the desired output?						
Show the answers to all the questions		Use the function and updated prompt on all the given questions to generate the responses.						
Provide comments/observations for all the answers received		How does the response change compared to the base LLM?						

Project Evaluation Rubric

1

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3

Vector Database Setup for RAG

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What to Cover?

Provide a brief overview of the dataset

Explain the parameters used to split the data (`chunk_overlap`, `chunk_size`) and why the selected values are optimal

Mention the name of the embedding model used

Guiding Questions and Considerations

What is the source and format of the data?
What is the total volume and diversity of the content?

What are the considerations for choosing chunk size and chunk overlap?

How to choose an embedding model?

Project Evaluation Rubric

1	2	3	4	Answering Questions with the Base RAG System	6	7	8	9
What to Cover?				Guiding Questions and Considerations				
Mention and explain the system prompt				What instructions define the AI's persona and rules for using the context?				
Mention and explain the user prompt				How is the retrieved context and the user's query structured together?				
Show the answers to all the questions				Define the response generation function. Get the responses for all the questions provided.				
Provide comments/observations for all the answers received				How does the response change on using RAG?				

Project Evaluation Rubric

1	2	3	4	5	Answering Questions with the Tuned RAG System	7	8	9

Project Evaluation Rubric

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Output Evaluation

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What to Cover?

Choose the LLM for evaluation
with proper rationale

Define evaluation parameters

Guiding Questions and Considerations

What are the considerations when selecting an evaluator LLM?

What metrics can be used to objectively evaluate for response accuracy
and hallucination?

Project Evaluation Rubric

1	2	3	4	5	6	Output Evaluation	8	9
What to Cover? Evaluate responses across all 4 techniques applied for the given set of questions Provide insightful comments on the evaluation outcomes, and any performance differences observed across the methods					Guiding Questions and Considerations How do the evaluation metrics compare numerically across the four techniques? What specific patterns or failure modes explain the performance gap between the techniques?			

Project Evaluation Rubric

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**Business Insights &
Recommendations**

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What to Cover?

Conclude with the key takeaways
for the business

Guiding Questions and Considerations

Are the key insights from modeling included in final takeaways, along
with comments on their alignment to the business objective?

Logical next steps are a key inclusion.

Project Evaluation Rubric

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Presentation / Notebook -
Overall Quality

Low-Code Considerations

Structure and flow - everything sits well in a story

Crispness - focus on key points

Visual appeal - layout supports readability

Conclusion and Business Recommendations

Full-Code Considerations

Clear structure and flow

Well-commented and executed code

Conclusion and Business Recommendations

Project Submission Guidelines: Low-Code

For learners who aspire to be in managerial roles in the future, focusing on solution review, interpretation, recommendations, and communication with business stakeholders

Download the dataset and the *Template Notebook: Low-Code Version* (this is a template notebook)

Fill in the blanks in the notebook to complete and execute the code to solve the questions and perform all the tasks as per the grading rubric

Once the notebook is completely executed and necessary outputs obtained, a business presentation (using Microsoft PowerPoint, Google Slides, etc.) has to be created

Project Submission Guidelines: Low-Code

The presentation should contain observations, insights, and recommendations for the business problem

The presentation template provided can be referred to as a sample

Once the presentation is complete, convert the presentation to .pdf format

The presentation should be submitted as a PDF file (.pdf) and NOT as a .pptx file

Please make sure that all the sections mentioned in the grading rubric have been covered in the submission

Project Submission Guidelines: Full-Code

For learners who aspire to be in hands-on coding roles in the future, focusing on building solution from scratch

Download the dataset and the *Template Notebook: Full-Code Version* (this is a template notebook containing high-level steps to perform and insight-based questions)

Write necessary code to solve the questions and perform all the tasks as per the grading rubric

Clearly write down observations, insights, and recommendations for the business problem based on the analysis

Once the notebook is complete, download it as a **.ipynb** file and convert it to a **.html** file

Project Submission Guidelines: Full-Code

The notebook should be submitted as an HTML file (.html) and NOT as a notebook file (.ipynb)

The conversion can be done via [free online tools](#)

Please make sure that all the sections mentioned in the grading rubric have been covered in the submission



Why is the vector database is taking longer than expected to be created?

Chunk size and chunk overlap control the time and space complexity of the vector database.

Lowering the chunk size will help reduce the memory needed to store these chunks.

Lowering the chunk_overlap parameter will reduce the number of generated chunks.

Does a higher k value always mean a better answer?

Not necessarily. While a higher k (more chunks) provides more information, it can also introduce "noise" or irrelevant data that confuses the model, leading to less precise answers.

In a basic RAG system, k is your retrieval limit. When you have a dataset spread across multiple files, choosing k is a balancing act between coverage and clarity.

If you set k too high, you risk overflowing the context window.

What is the difference between Faithfulness and Relevancy?

Faithfulness (The "No Hallucination" Metric)

Faithfulness measures the relationship between the Generated Answer and the Retrieved Context.

- The Core Question: Did the LLM make this up, or did it get it from the PDFs?
- The Goal: To ensure the model is "faithful" to the source material. If the model uses its own pre-trained knowledge to answer a question that wasn't in your documents, it is considered "unfaithful," even if the answer is technically true in the real world.

Answer Relevancy (The "Directness" Metric)

Relevancy measures the relationship between the Generated Answer and the User's Query.

- The Core Question: Does this response actually address what the user asked?
- The Goal: To ensure the model doesn't "ramble" or provide a factually correct but useless answer. An answer can be 100% faithful to the text (meaning every word came from the PDF) but still be irrelevant if it doesn't answer the specific prompt.



Power Ahead!

